

Cardell Taylor

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EDUCATION

Grand Valley State University

Bachelor of Science in Computer Science, Minor in Mathematics

August 2022 - May 2026

Allendale, MI

EXPERIENCE

AI & ML Solutions Engineer - Contract

ScalableAI

February 2026 – Present

Grand Rapids, MI

- Engineered a manufacturing cost estimation pipeline for a company's die casting quote archive, extracting relevant manufacturing complexity features.
- Developed and benchmarked regression model architectures (XGBoost, Ridge, Lasso, etc) on a 700-sample quote dataset, achieving $R^2=0.84$ with per-part pricing and MAPE as the primary evaluation metric
- Built a full-stack RFQ quoting web interface (JS + FastAPI) to interact with the prediction model
- Developed a RAG engine for quick semantic searching over dense service manuals

ML Research Assistant

Grand Valley State University

August 2023 – December 2025

Allendale MI

- Developed data pipelines to handle ~30M sample scRNA-seq dataset from the Chan Zuckerberg Initiative
- Designed a model to integrate and generate genomics data using a multi-modal VAE architecture
- Utilized SLURM to manage jobs on university's HPC cluster
- Generated publication-quality figures for research presentations

PROJECTS

SecondMind | *Python, LangChain, ChromaDB*

November 2025

- Built a local RAG notes assistant using Python, LangChain, and ChromaDB, enabling semantic search over a personal markdown knowledge base with no cloud dependencies
- Integrated sentence-transformer embeddings (all-MiniLM-L6-v2) with a persistent vector database and a locally-hosted LLM (Ollama / llama3.2) for full privacy preservation

MNIST Michi-GAN | *Python, Pytorch*

August 2024

- Exploration of the MichiGAN framework for generating disentangled representations of the MNIST dataset, as detailed in the paper *MichiGAN: sampling from disentangled representations of single-cell data using generative adversarial networks*
- Evaluated performance using common metrics like Fréchet Inception Distance (FID) and Inception Score

Autoencoder | *C++, OpenMP, Eigen*

Sep 2023 - Nov 2023

- Implemented a neural network from scratch that can be trained to compress and reconstruct image data
- Implements ADAM, a modern optimization technique reduce model convergence time
- Used multithreading with OpenMP to boost performance
- Later moved to python to test adversarial feedback techniques

GRANTS/AWARDS

Student Summer Scholar Grant

Grand Valley State University

March 2024

- Grant awarded to undergraduate students to support promising projects throughout the summer semester

TECHNICAL SKILLS

Languages: Python, C++, Java, C#, OCaml, Rust

Frameworks/Libraries: Numpy, Pytorch, Keras, Tensorflow, OpenGL, Eigen, scikit-learn, Pandas, Seaborn

Developer Tools: Git, VS Code, PyCharm, Eclipse, CMake, Premake, AWS, SageMaker, EC2, Docker